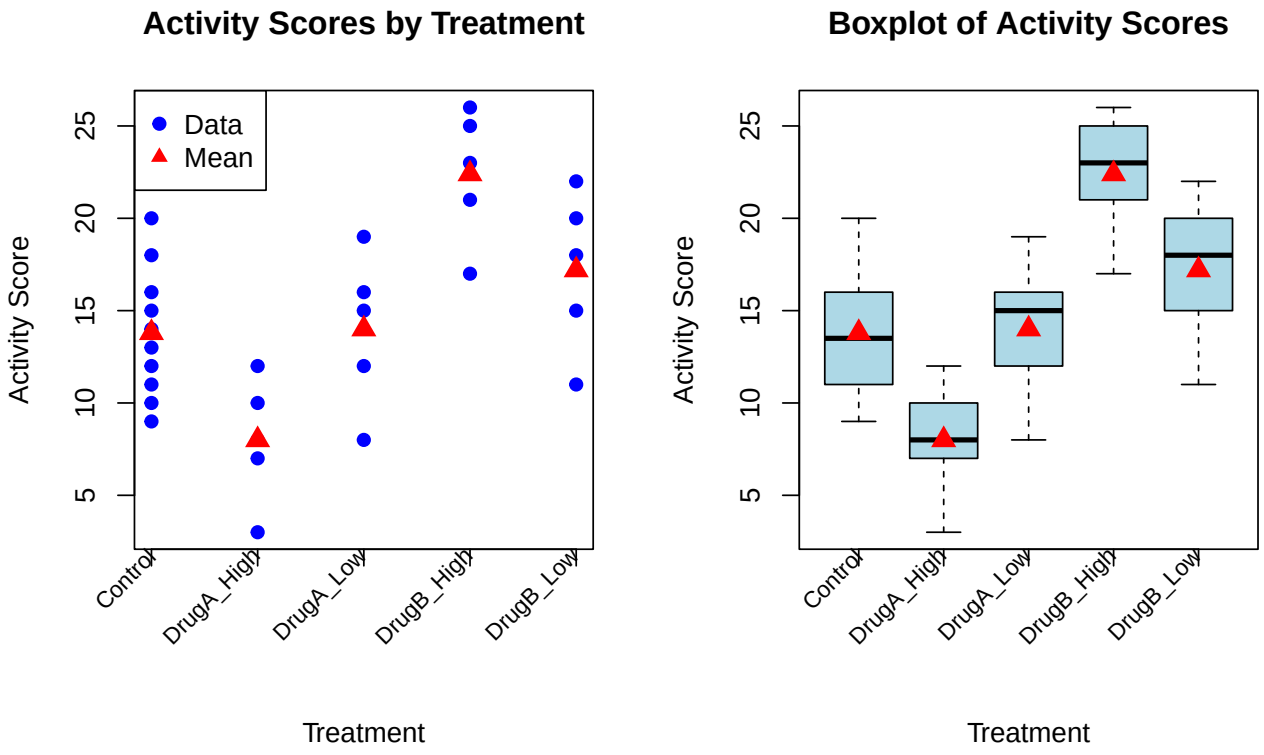
STAT 5080 HW4 Solutions

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Problem 1: Drug Effects on Rat Activity

Part (a): Graphical Comparison of Treatment Means



Conclusions from the graph:

- The Control group shows moderate activity (mean 13.8)
- Drug A Low dosage shows similar activity to control
- Drug A High dosage shows lower activity (depressant effect), with mean around 8
- Drug B groups (both Low and High) show higher activity (stimulant effect)
- Drug B High dosage shows the highest activity level (mean 22.4)
- There appears to be clear differences among the five treatment groups

Part (b): One-Way ANOVA

Hypotheses:

- $H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = \tau_5 = 0$ (all treatment effects are equal)
- H_a : At least one $\tau_i \neq 0$ (at least one treatment effect differs)

or equivalently:

- $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ (all treatment means are equal)
- H_a : At least two treatment means differ

One-Way ANOVA Table:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treatment	4	561.8667	140.4667	9.9312	1e-04
Residuals	25	353.6000	14.1440	NA	NA

```
##
## MSE = 14.144
## F-statistic = 9.9312
## p-value = 5.9427e-05
```

Conclusion:

With p-value < 0.05, we reject H_0 and conclude that there are significant differences among the five treatment groups. This is consistent with the graphical analysis in part (a), which showed clear visual differences among treatment means, particularly between the control, Drug A high, and Drug B high groups.

Part (c): Contrast Coefficients

	Control	DrugA_Low	DrugA_High	DrugB_Low	DrugB_High
Control vs Experimental	4	-1	-1	-1	-1
Drug A vs Drug B	0	1	1	-1	-1
Low vs High (Drug A)	0	1	-1	0	0
Low vs High (Drug B)	0	0	0	1	-1

Part (d): Orthogonality Check

Yes, the four comparisons in (c) are mutually orthogonal.

Part (e): Sum of Squares and Significance Tests

```
##
## Contrast Analysis:
```

Contrast	Estimate	SS	MS	F_value	p_value
Control vs Experimental	-6.4	17.0667	17.0667	1.2066	0.2825
Drug A vs Drug B	-17.6	387.2000	387.2000	27.3756	0.0000
Low vs High (Drug A)	-6.0	90.0000	90.0000	6.3631	0.0184
Low vs High (Drug B)	5.2	67.6000	67.6000	4.7794	0.0384

```
##
## **Significance at  = 0.05:**
## Control vs Experimental : Not Significant (p = 0.2825 )
## Drug A vs Drug B : Significant (p = 0 )
## Low vs High (Drug A) : Significant (p = 0.0184 )
## Low vs High (Drug B) : Significant (p = 0.0384 )
```

Part (f): Bonferroni Multiple Comparisons (Planned)

Bonferroni Confidence Intervals ($\alpha = 0.05, m = 4$)

Contrast	H0	Lower_Limit	Upper_Limit	Conclusion
Control vs Experimental	Control vs Experimental = 0	-22.0818	9.2818	No significant difference (CI contains 0)
Drug A vs Drug B	Drug A vs Drug B = 0	-26.6539	-8.5461	Negative difference is significant
Low vs High (Drug A)	Low vs High (Drug A) = 0	-12.4021	0.4021	No significant difference (CI contains 0)
Low vs High (Drug B)	Low vs High (Drug B) = 0	-1.2021	11.6021	No significant difference (CI contains 0)

Part (g): Scheffe Multiple Comparisons (Not Planned)

Scheffe Confidence Intervals ($\alpha = 0.05$)

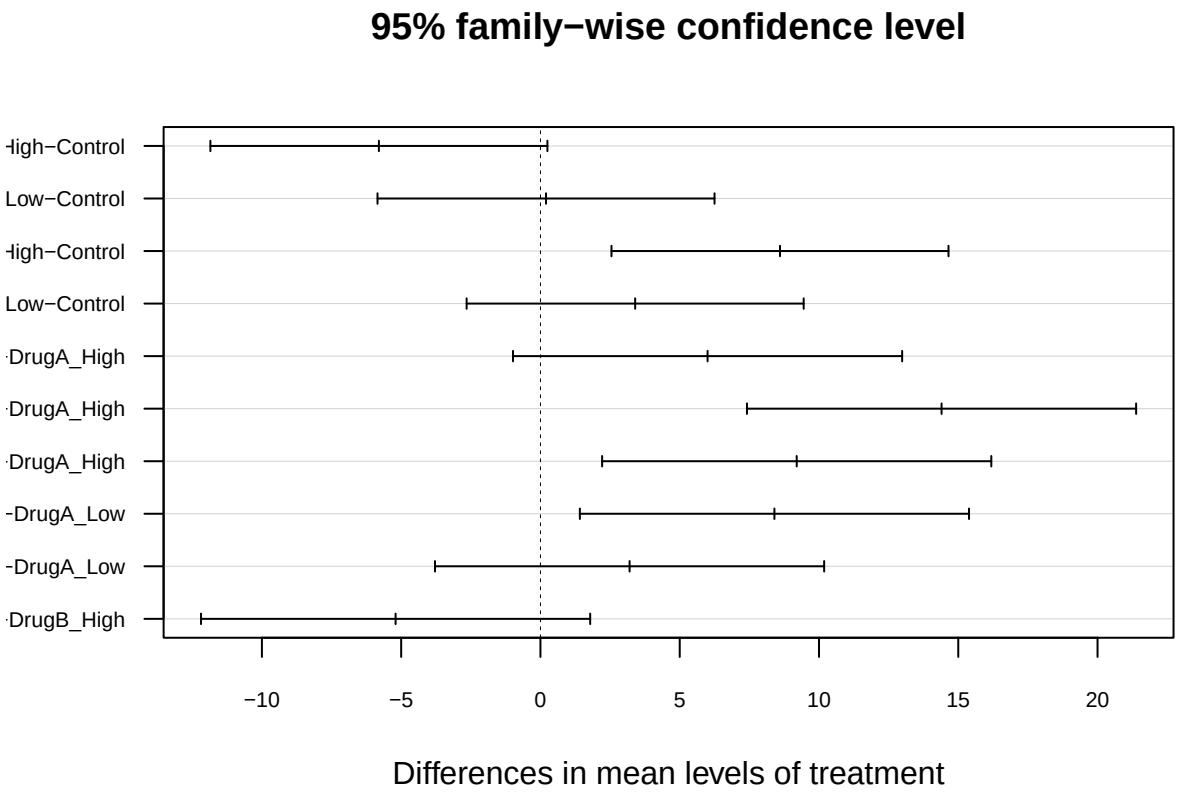
Contrast	H0	Lower_Limit	Upper_Limit	Conclusion
Control vs Experimental	Control vs Experimental = 0	-25.7542	12.9542	No significant difference (CI contains 0)
Drug A vs Drug B	Drug A vs Drug B = 0	-28.7741	-6.4259	Negative difference is significant
Low vs High (Drug A)	Low vs High (Drug A) = 0	-13.9013	1.9013	No significant difference (CI contains 0)
Low vs High (Drug B)	Low vs High (Drug B) = 0	-2.7013	13.1013	No significant difference (CI contains 0)

Part (h): Dunnett's Test (Control vs Each Treatment)

```
##
## Dunnett's Test: Control vs Each Treatment (  = 0.05)
##
##   Simultaneous Tests for General Linear Hypotheses
##
## Multiple Comparisons of Means: Dunnett Contrasts
##
##
## Fit: lm(formula = activity ~ treatment, data = rat_data)
##
## Linear Hypotheses:
##                            Estimate Std. Error t value Pr(>|t|)
## DrugA_High - Control == 0    -5.80       2.06  -2.816  0.03425 *
## DrugA_Low - Control == 0      0.20       2.06   0.097  0.99995
## DrugB_High - Control == 0     8.60       2.06   4.175  0.00123 **
## DrugB_Low - Control == 0      3.40       2.06   1.651  0.33812
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## (Adjusted p values reported -- single-step method)
##
```

```
##
## Dunnett's Confidence Intervals:
##
##   Simultaneous Confidence Intervals
##
## Multiple Comparisons of Means: Dunnett Contrasts
##
##
## Fit: lm(formula = activity ~ treatment, data = rat_data)
##
## Quantile = 2.6437
## 95% family-wise confidence level
##
##
## Linear Hypotheses:
##                            Estimate lwr      upr
## DrugA_High - Control == 0  -5.8000 -11.2458  -0.3542
## DrugA_Low - Control == 0    0.2000  -5.2458   5.6458
## DrugB_High - Control == 0   8.6000   3.1542  14.0458
## DrugB_Low - Control == 0    3.4000  -2.0458   8.8458
```

Part (i): Tukey's HSD Test




```
##
## Tukey's HSD Test (  = 0.05):
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
```

```
##
## Fit: aov(formula = activity ~ treatment, data = rat_data)
##
## $treatment
##                          diff         lwr        upr     p adj
## DrugA_High-Control     -5.8 -11.8496696  0.2496696 0.0647914
## DrugA_Low-Control       0.2  -5.8496696  6.2496696 0.9999784
## DrugB_High-Control      8.6   2.5503304 14.6496696 0.0026704
## DrugB_Low-Control       3.4  -2.6496696  9.4496696 0.4808852
## DrugA_Low-DrugA_High    6.0  -0.9855567 12.9855567 0.1174701
## DrugB_High-DrugA_High  14.4   7.4144433 21.3855567 0.0000232
## DrugB_Low-DrugA_High    9.2   2.2144433 16.1855567 0.0057120
## DrugB_High-DrugA_Low    8.4   1.4144433 15.3855567 0.0128768
## DrugB_Low-DrugA_Low     3.2  -3.7855567 10.1855567 0.6664957
## DrugB_Low-DrugB_High   -5.2 -12.1855567  1.7855567 0.2175353
```

Recommendation for High Activity: Based on Tukey's test, Drug B High dosage has the highest mean activity score (mean = 22.4) and is significantly different from Control and Drug A High. Therefore, if high activity is desired, **Drug B High dosage is recommended**.

Problem 2: Vitamin Study Design

Part (a): Top 5 Comparisons of Interest

For this study with 5 groups (Control, Brand A Low, Brand A High, Brand B Low, Brand B High), here are the top 5 comparisons:

Comparison	Point Estimate	Standard Error	Reason of Interest
1. Control vs Experimental	$4\bar{y}_1 - \bar{y}_2 - \bar{y}_3 - \bar{y}_4 - \bar{y}_5$	$\sqrt{MSE}$	Test if any vitamin treatment differs from placebo
2. Brand A vs Brand B	$\bar{y}_2 + \bar{y}_3 - \bar{y}_4 - \bar{y}_5$	$\sqrt{0.2 \times MSE}$	Test overall difference between Brands
3. Low vs High Dose	$\bar{y}_2 - \bar{y}_3 + \bar{y}_4 - \bar{y}_5$	$\sqrt{0.2 \times MSE}$	Test overall difference between Doses
4. Low vs High (Brand A)	$\bar{y}_2 - \bar{y}_3$	$\sqrt{0.1 \times MSE}$	Test simple effect of dose within Brand A
5. Low vs High (Brand B)	$\bar{y}_4 - \bar{y}_5$	$\sqrt{0.1 \times MSE}$	Test simple effect of dose within Brand B

Part (b): Orthogonality Check

Orthogonality Check for Top 5 Contrasts

$v_1 \cdot v_2 = 0$
$v_1 \cdot v_3 = 0$
$v_1 \cdot v_4 = 0$
$v_1 \cdot v_5 = 0$
$v_2 \cdot v_3 = 0$
$v_2 \cdot v_4 = 0$
$v_2 \cdot v_5 = 0$
$v_3 \cdot v_4 = 2$

$v_3 \cdot v_5 = 2$
$v_4 \cdot v_5 = 0$

Answer: NO. The five contrasts are not mutually orthogonal.
For contrasts to be mutually orthogonal, all pairwise dot products must equal 0. Here, two dot products are non-zero.

Part (c): Most Interesting Comparison and 95% CI

Let the most interesting contrast be defined as:

$$C = 4\bar{y}_1 - \bar{y}_2 - \bar{y}_3 - \bar{y}_4 - \bar{y}_5$$

The general formula for the 95% Confidence Interval is:

$$C \pm t_{(N-a,\,0.975)} \times \sqrt{MSE \times \sum \frac{c_i^2}{n_i}}$$

Where:

- $N = 100$ (total sample size)
- $a = 5$ (number of groups)
- $df = N - a = 95$
- $t_{(95,\,0.975)} \approx 1.985$
- MSE = Mean Square Error from ANOVA
- $\sum \frac{c_i^2}{n_i} = \frac{16}{20} + \frac{1}{20} + \frac{1}{20} + \frac{1}{20} + \frac{1}{20} = \frac{20}{20} = 1$

Therefore, the 95% CI is:

$$(4\bar{y}_1 - \bar{y}_2 - \bar{y}_3 - \bar{y}_4 - \bar{y}_5) \pm 1.985 \times \sqrt{MSE}$$